

Screening for Biliary Atresia: Stool Colour Cards and Early Direct Bilirubin.

Imagine two newborn girls born on the same day. Both have biliary atresia, a disease in which the bile ducts outside the liver become inflamed, scarred and finally blocked. The first girl lives in a place where nobody looks for the disease until someone notices that she is still yellow. Her parents are told that breast milk jaundice is common, her stools look a little pale at six weeks, and she has her operation at about eight weeks of age. The second girl has a small blood test on her second day of life, the same blood sample that was already being taken to check her total bilirubin. The laboratory notices that her conjugated bilirubin is slightly above the normal range, the test is repeated at two weeks, and she has her operation at four weeks. The two girls have the same disease, but they do not have the same chance of keeping their own liver.

The clinical question of this chapter is whether we can find biliary atresia before it becomes visible, and, if we can, which tool a health system should use. The main idea is this. Biliary atresia changes the blood from the first days of life, but it changes the colour of the stool only later. A stool colour card, which asks parents to compare their baby's stool with printed colours, is cheap and has improved outcomes in Taiwan, but it detects a late sign and its sensitivity varies from about one half to almost all cases depending on how the programme is run. Screening with a direct or conjugated bilirubin in the first sixty hours of life, repeated at two weeks, has so far found every case in the published cohorts and has moved the operation about three weeks earlier, but it costs more and depends completely on a recall system. Neither tool works unless someone acts on the result the same day.

We will move through the topic in order. First, the biology: why the blood is abnormal before the stool. Second, the story of the stool colour card in Japan, Taiwan, Canada and elsewhere. Third, the newer evidence on early bilirubin screening from Texas and Utah. Fourth, a direct comparison of the two tools, their failure points, and their cost. Fifth, what the guidelines say. We finish with what to do differently in clinic and some questions to test yourself.

A note on sources. The main documents are the 2017 guideline on cholestatic jaundice in infants from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, written by Fawaz and colleagues; the 2025 clinical report of the American Academy of Pediatrics, the AAP, on identifying infants with biliary atresia by two to four weeks of life, written by Harpavat and colleagues; and the 2026 external review for the United Kingdom National Screening Committee. Neither the American Association for the Study of Liver Diseases nor the European Association for the Study of the Liver has a statement on newborn screening for this disease.

Part one. Why the clock matters, and why the blood speaks first.

The treatment for biliary atresia is the Kasai portoenterostomy, an operation in which the surgeon removes the scarred ducts at the porta hepatis and sews a loop of bowel onto the cut surface of the liver, so that bile can drain from the tiny ductules that are still open. The operation works best when it is done early, before the liver has become cirrhotic. In the French national series of six hundred and ninety-five children published by Serinet and colleagues in 2009, survival with the native liver fell step by step with each fifteen-day delay in the age at surgery, and the authors estimated that operating on every child before forty-six days of age would spare about six percent of all paediatric liver transplants in France each year. Screening does not change the disease. It only changes the date of the operation. So every screening programme has to be judged by how many days it buys, at what price, and how many healthy babies it worries along the way.

Now consider what each screening tool actually detects. A stool colour card detects acholia, which means that bile pigment is no longer reaching the gut. That happens only when the obstruction has become nearly complete. In some infants with biliary atresia the stool is still yellow or light yellow at two, three, or even four weeks, and it pales later. So the card waits for a late sign.

Conjugated bilirubin behaves differently. In a healthy newborn, conjugated bilirubin is very low, because the liver's capacity to export bile is good even when its capacity to conjugate is still immature. When bile cannot leave the liver, conjugated bilirubin leaks back into the blood almost at once. In 2011, Harpavat, Finegold and Karpen in Houston looked back at sixty-one infants with biliary atresia treated at Texas Children's Hospital between 2007 and 2010. Thirty-four of them happened to have had a direct or conjugated bilirubin measured in the first ninety-six hours of life, as part of routine newborn care. All thirty-four values were above the upper limit of normal. Between twenty-four and forty-eight hours of age, the average direct bilirubin was about one point four milligrams per decilitre in the infants with atresia, compared with about zero point two in healthy controls, and the lowest value in an atresia infant, zero point eight, was higher than the highest value in any control, which was zero point seven. Seventy-nine percent of these infants had a direct bilirubin that was twenty percent or less of the total, so the old percentage rule would have called them normal. And only about half of them received any follow-up of the abnormal number.

The message is that biliary atresia is a disease that is already present at birth or very soon after, at least in its effect on bile flow. The blood shows it on the first or second day. The stool shows it weeks later. This is the biological reason why blood screening can be both earlier and more sensitive than a card.

Part two. The stool colour card in Japan.

The stool colour card began in Japan. Matsui and Dodoriki described the idea in the mid-1990s, and Tochigi Prefecture used a card at the one-month health check from 1994. In 2015, Gu and colleagues, writing in the *Journal of Pediatrics*, reported nineteen years of results from Tochigi, covering three hundred and thirteen thousand two hundred and thirty infants born between 1994 and 2011. There were thirty-four cases of biliary atresia, an incidence of about one in nine thousand. At the one-month check, the card had a sensitivity of seventy-six and a half percent, a

specificity of ninety-nine point nine percent, and a positive predictive value of about thirteen percent, which means that roughly one in eight infants with an abnormal card had atresia. The mean age at Kasai was about sixty days. Survival with the native liver was about eighty-eight percent at five years, seventy-seven percent at ten years, and forty-nine percent at fifteen years. Japan then placed an updated card in the national maternal and child health handbook from 2012.

A later paper from the same group, published in 2020, is a useful warning. In Sapporo, between 2012 and 2015, the card was used for more than thirty-seven thousand infants. There were two infants with biliary atresia, and both still had normal-looking stools at the one-month check. The sensitivity at that visit was therefore zero. A card checked once, at one month, will miss the infant whose stools turn pale at six weeks. The card must be used continuously by parents, not just once by a nurse.

Part three. Taiwan, the flagship programme.

Taiwan built the best-known national programme. A seven-colour card was printed into the child health booklet that every family receives, and parents were asked to compare the stool with the card, especially at the one-month visit. A regional pilot ran from 2002 to 2003, and Chen and colleagues reported it in *Pediatrics* in 2006. About seventy-eight thousand infants took part. The card identified twenty-six of twenty-nine infants with atresia before sixty days of age, a sensitivity of almost ninety percent, and the positive predictive value was about twenty-nine percent, much higher than in Europe because biliary atresia is about three times as common in Taiwan. Fifty-nine percent of infants had their Kasai before sixty days.

The card went national in 2004, and an abnormal colour had to be reported within twenty-four hours. Hsiao and colleagues reported the first two national years in *Hepatology* in 2008. The incidence was about one point eight per ten thousand births. In the first national year the card detected only seventy-three percent of cases before sixty days. In the second year it detected ninety-seven percent. Over the same two years, the proportion of infants who had their Kasai before sixty days rose from sixty percent to seventy-four percent. Three months after the Kasai, about sixty percent of infants had cleared their jaundice, compared with thirty-seven percent in the years before screening. The lesson is that a national programme takes time to mature: families, nurses and paediatricians had to learn to use the card.

The outcome data came in 2011, when Lien and colleagues compared eighty-nine infants with biliary atresia from the years before screening, 1990 to 2000, with infants from the pilot and national screening cohorts. Kasai before sixty days rose from about forty-nine percent to sixty-six percent. Jaundice clearance three months after surgery rose from thirty-five percent to sixty-one percent. Three-year survival with the native liver and without jaundice rose from thirty-two percent to fifty-seven percent. Five-year survival with the native liver and without jaundice, which could only be measured in the earlier pilot cohort, rose from twenty-seven percent to sixty-four percent, and five-year overall survival rose from fifty-six percent to eighty-nine percent. All these differences were statistically significant. On a chart, the grey bars for the years before screening sit

well below the red bars for the screening years in every one of the four outcomes.

It is important to read these numbers carefully. This was a comparison of two eras, not a randomised trial. Over the same decade, surgery improved, care was concentrated in experienced centres, and liver transplantation became more available. Some of the gain belongs to the card, and some belongs to everything else.

National insurance data give a more modest picture. Tseng and colleagues, also in Pediatrics in 2011, analysed claims from the whole of Taiwan between 1996 and 2008. The median age at Kasai fell only from fifty-one to forty-eight days, and the proportion of infants operated within sixty days rose from about sixty-nine percent to seventy-four percent. Very late referrals fell by about half. In 2013, Chiu and colleagues showed that the card worked as well in preterm infants as in term infants, with a sensitivity of about ninety-six percent, but only forty-four percent of preterm infants had their Kasai before sixty days, compared with sixty-nine percent of term infants. Sick preterm infants have many other reasons for cholestasis, such as parenteral nutrition and sepsis, and atresia is often recognised late. A later national cohort reported that mortality fell from about twenty-six percent to sixteen percent, although the transplant rate stayed at about thirty percent. The overall message from Taiwan is that the card finds the infant, but the referral pathway decides the date of surgery.

Part four. Canada and other programmes.

British Columbia in Canada is the best example of a card programme in a low-prevalence Western population. Before the programme, Schreiber and colleagues studied about six thousand families and modelled the costs, and they estimated that passive distribution of the card would cost about twenty-two thousand Canadian dollars per life-year gained. The province then distributed the card through all one hundred and twenty-six maternity units. Woolfson and colleagues reported the first two years, from April 2014 to March 2016, in the Journal of Pediatric Gastroenterology and Nutrition in 2018. There were eighty-seven thousand five hundred and eighty-three births and six infants with biliary atresia. The card reached ninety-four percent of families, but it flagged only three of the six cases, so the sensitivity was fifty percent. The positive predictive value was about four percent. When the card worked, the median age at Kasai was forty-nine days. When it failed, the median age was one hundred and sixteen days. The programme cost about thirty thousand Canadian dollars a year, or sixty-eight cents per birth. This is the essence of the stool colour card in a low-prevalence setting: very cheap, easy to distribute, and with a sensitivity that can be no better than a coin toss unless the programme is actively audited.

Other countries tell similar stories. Switzerland started a national card campaign in 2005 and added the card to the national child health booklet in 2019. Its accuracy has not been published, but unpublished data reported by Wildhaber and Calinescu in 2025 suggest that the median age at Kasai fell from sixty-eight days to fifty-seven days. In Beijing, a digital version of the card was used for about thirty thousand infants, and it detected the two infants with atresia only at fifty-two and fifty-five days of age. In 2024, Gopal and colleagues published a meta-analysis in PLOS One.

Across seven card studies including almost one million infants, the pooled sensitivity was about eighty percent, with a confidence interval from about seventy-one to eighty-six percent, and the specificity was ninety-nine point nine percent. Sensitivity depends on prevalence, on when the card is read, and on whether anyone acts on an abnormal colour within a day.

Part five. Early direct bilirubin screening.

The Houston observation from 2011 led to a prospective test. In a letter in the New England Journal of Medicine in 2016, Harpavat and colleagues described a pilot of eleven thousand six hundred and thirty-six newborns in four hospitals. About one hundred and twenty had a raised direct or conjugated bilirubin on the first sample, eleven were still raised at about two weeks, and among those eleven were two infants with biliary atresia, one with alpha-one antitrypsin deficiency, and one with Rh disease.

The definitive study was published in the Journal of the American Medical Association, JAMA, in 2020. Harpavat and colleagues screened one hundred and twenty-four thousand three hundred and eighty-five newborns in fourteen hospitals in south Texas between 2015 and 2018. The protocol had two stages. In stage one, before sixty hours of age, a direct or conjugated bilirubin was measured on the blood sample that was already being taken, and the result was called positive if it was above the hospital's own reference range. For conjugated bilirubin the upper limit was only zero point two milligrams per decilitre. About one percent of newborns screened positive at this stage. In stage two, by two weeks of age, those infants had a repeat test, and it was called positive if the value was higher than the first one or above one milligram per decilitre. Only about one in a thousand newborns was positive at this second stage.

There were seven infants with biliary atresia, and the screen found all seven. The sensitivity was therefore one hundred percent, although with only seven cases the ninety-five percent confidence interval runs from fifty-six to one hundred percent. The specificity was ninety-nine point nine percent and the positive predictive value about six percent. Among the one hundred and twelve false positives, about eight percent turned out to have another cholestatic liver disease, so the screen also finds children who need a hepatologist for other reasons.

The effect on the clock was large. At the screening centre, the authors compared twenty-four infants with atresia treated before screening with nineteen treated after it. The mean age at Kasai fell from fifty-six days to thirty-six days. The proportion of infants operated by thirty days of age rose from about thirteen percent to fifty-eight percent. The proportion whose bilirubin normalised within three months of surgery rose from forty-two percent to seventy-nine percent.

Transplant-free survival at one year rose from seventy-one percent to ninety-five percent, but this last difference did not reach statistical significance, with a P value of zero point zero six. So the screen clearly moves the operation earlier and improves early bile drainage, but it has not yet been proven to save livers in the long term.

Other cohorts support the signal and expose the logistics. In Utah, Kastenbergh and colleagues reviewed more than two hundred and fifty thousand newborns between 2005 and 2019. All

twenty-six infants with biliary atresia had a raised direct bilirubin in the newborn period, but so did about one in seventy-five other newborns, and the median age at Kasai was still about seventy days. The numbers were in the chart, but nobody acted on them. A later Utah implementation study screened ninety-nine and a half percent of newborns in its pilot hospitals, with a net false positive rate of about three in a thousand after retesting. In the 2024 meta-analysis, the pooled sensitivity of newborn direct or conjugated bilirubin across five studies and more than six hundred thousand infants was one hundred percent, with a specificity of about ninety-nine percent. The point is simple. A screen is a test plus a recall system. Without the recall, it is only data.

Part six. Where programmes fail, and how the two tools compare.

It helps to think of screening as a chain with five links. The first link is distribution: the card must be in every booklet, or the direct bilirubin must be in every newborn order set. The second is reading: parents must look at the stool against the card, or the laboratory must flag any value above its range. The third is reporting: Taiwan required a phone call within twenty-four hours, and Texas recalled positive infants at two weeks. The fourth is referral: the infant must be seen by a hepatologist the same day if the stool is pale or the conjugated bilirubin is one milligram per decilitre or more. The fifth is the operation itself, which should be done in an experienced centre. Most published failures are not failures of the test. They are a card that nobody looked at, a pale stool photographed but not reported, a raised value never repeated, or a sick preterm infant whose cholestasis was blamed on parenteral nutrition.

When we put the two tools side by side, the comparison is clear. The card has a pooled sensitivity of about eighty percent, with a range from fifty to ninety-seven percent across programmes, while early bilirubin screening has found every case so far. The positive predictive value is low for both, about four to twenty-nine percent for the card depending on prevalence, and about six to eighteen percent for blood. The card usually raises the alarm at two to four weeks, the blood before sixty hours with confirmation by two weeks. The card costs less than a dollar per birth and needs no needle, while blood screening needs laboratory work, a second sample and staff to recall families. The weak spot of the card is late pallor, misreading and inaction. The weak spot of blood screening is the missed repeat sample, especially after early discharge. The two are complementary. Even where blood screening exists, the card still matters for the infant whose early values were normal, or whose repeat test was never done.

Part seven. What screening costs.

The economic models disagree, mostly because they make different assumptions about transplant costs, prevalence and outcomes. In the United States, Mogul and colleagues in 2015 modelled a twenty-year horizon and found that a universal stool card would actually save money: about eight and a half million dollars per birth cohort, with about thirty life-years gained and eleven fewer transplants. The card was dominant, meaning cheaper and better, in more than ninety-seven percent of simulations. In Canada, as we saw, the estimate was about twenty-two thousand Canadian dollars per life-year gained. In Japan, Hoshino and colleagues in 2023 modelled about

nine hundred and forty thousand newborns over twenty-five years. Compared with no screening, the card cost about three hundred and forty thousand dollars per event-free life-year. Blood screening compared with the card cost about five hundred and thirteen thousand dollars per event-free life-year, while preventing sixteen more transplants. Both figures are far above the Japanese willingness to pay of about thirty-three thousand dollars per quality-adjusted life-year. The summary is that the card wins on price in most settings, while blood screening wins on days and has not yet been shown to be cheap.

Part eight. What the guidelines say.

The 2017 NASPGHAN and ESPGHAN guideline does not make a formal recommendation on stool colour cards. It states that looking at the stool directly is essential, it cites the Taiwanese experience, and it notes that cards are not systematically used in North America or Europe. It defines a conjugated or direct bilirubin above one milligram per decilitre as pathological whatever the total.

The 2025 AAP clinical report goes further. It asks primary care clinicians to identify infants with biliary atresia by two to four weeks of life, so that the Kasai can be done before thirty days. It asks them to consider a direct or conjugated bilirubin in any infant who is jaundiced, and in any infant with pale stools at any age. It states that a newborn direct or conjugated bilirubin above the laboratory's reference range is abnormal even if it is above the limit by only zero point one milligrams per decilitre, and that such a value should be repeated rather than ignored. A value of one milligram per decilitre or more should lead to urgent referral to a paediatric gastroenterologist or hepatologist. It also suggests that a direct or conjugated bilirubin be added whenever a total bilirubin is drawn in a newborn. It does not mandate universal screening.

In the United Kingdom, the National Screening Committee reviewed the evidence again in 2026 and kept its position: population screening for biliary atresia is not recommended. The review found that card sensitivity ranged from zero to one hundred percent across studies, that the positive predictive value at United Kingdom prevalence would be only about four to six percent, and that there was not enough evidence for blood spot testing either. It also noted that the median age at Kasai in England and Wales, about forty-eight to fifty-one days, is similar to that achieved in some countries that screen. The National Institute for Health and Care Excellence, NICE, recommends measuring conjugated bilirubin in prolonged jaundice and seeking expert advice above twenty-five micromoles per litre. A 2025 Chinese guideline is more enthusiastic: it says the stool card may help, and that blood screening is promising, with a first test within sixty hours and a second within two weeks.

When we compare the documents, the action at a conjugated bilirubin of one milligram per decilitre or more is aligned: refer urgently. On stool cards and on universal blood screening, North America and Europe differ, but neither side mandates screening. The material difference is the target. North America now aims for a Kasai before thirty days, and asks clinicians to act on any value above the laboratory range. The United Kingdom accepts a national median near fifty days

and has no instruction about small elevations in the first days of life. For an individual infant, that difference can mean three weeks.

Part nine. What to do differently in clinic.

Here are the habits to take away. When you look at a newborn chart, look for a direct or conjugated bilirubin, and if it is above the laboratory range, even slightly, do not file it as normal; arrange a repeat by the two-week visit. If the repeat is one milligram per decilitre or more, refer to hepatology the same day, and do not be reassured by yellow stools, because the stool pales late. Give every family a stool colour card or a good photograph-based tool, and tell them to call the same day if the stool matches one of the pale colours, at any age. If a breastfed baby is still jaundiced at three weeks, fractionate the bilirubin even if the card looks normal, because the card misses about one in five cases or more. In the neonatal unit, do not let parenteral nutrition explain away a conjugated bilirubin until you have looked at the stool and the ultrasound. And if you help to design a regional programme, remember that the test is the easy part; the audit of distribution, reporting and referral is what makes it work.

Questions to test yourself.

Question one. Why is the blood abnormal before the stool in biliary atresia? Answer: because conjugated bilirubin leaks back into the blood as soon as bile flow is impaired, while the stool becomes pale only when the obstruction is nearly complete. In the Houston study, all thirty-four infants with atresia who had a direct or conjugated bilirubin in the first ninety-six hours had a value above normal.

Question two. What happened to card sensitivity in the first two years of the Taiwanese national programme? Answer: it rose from about seventy-three percent in 2004 to ninety-seven percent in 2005, and Kasai before sixty days rose from sixty to seventy-four percent.

Question three. What was the card's sensitivity in British Columbia, and what did a missed case cost? Answer: fifty percent, three of six cases. The median age at Kasai was forty-nine days when the card worked and one hundred and sixteen days when it failed.

Question four. Describe the two stages of the Texas screening protocol. Answer: a direct or conjugated bilirubin before sixty hours, positive if above the hospital reference range, and a repeat by two weeks, positive if higher than the first value or above one milligram per decilitre.

Question five. What did the Texas screen achieve? Answer: it found seven of seven infants with atresia, with a specificity of ninety-nine point nine percent and a positive predictive value of about six percent, and it lowered the mean age at Kasai from fifty-six to thirty-six days.

Question six. Why did Utah infants with raised newborn values still have a late Kasai? Answer: because the values were measured but nobody acted on them. There was a test but no recall system.

Question seven. What does the 2025 AAP clinical report say about a newborn direct bilirubin that is just above the laboratory range? Answer: it is abnormal, even by zero point one milligrams per

decilitre, and should be repeated; a value of one or more calls for urgent referral.

Question eight. What is the current position of the United Kingdom National Screening Committee? Answer: after its 2026 review it does not recommend population screening for biliary atresia, either with stool cards or with blood tests.

End of chapter.